

Component Level Events

Communication with child->parent

Sheetal Chougule

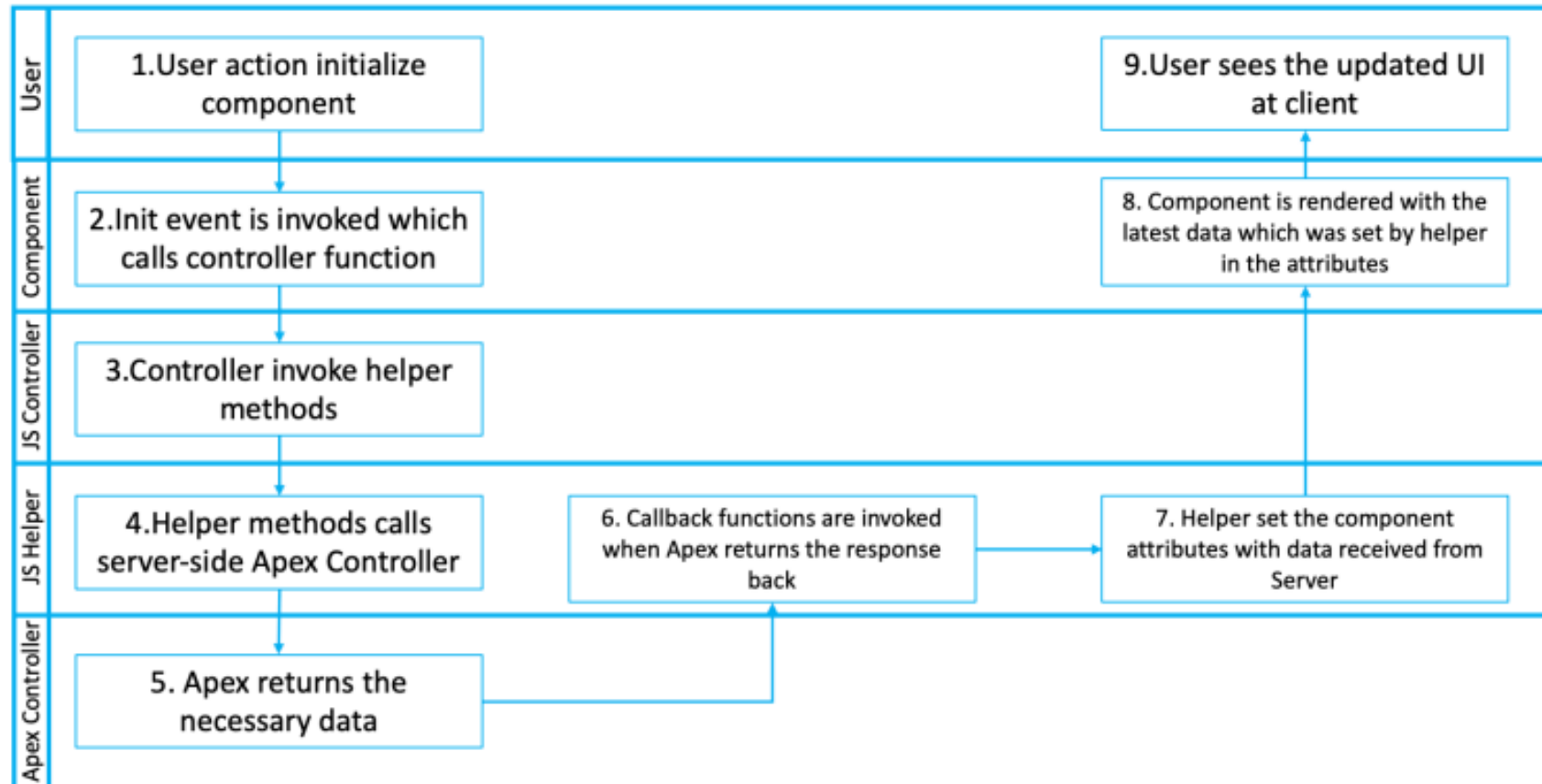




Mastering Component Events in Aura



Aura Component Lifecycle



What are Aura Events?

Aura events are communication mechanisms between components. There are Three types of Events are supported:

- ✓ Component Event
- ✓ Application Event
- ✓ System Event

Type	Direction	Use Case
Component	Child → Parent	Notify parent when something changes in child. When components are nested (child > parent) and When action originates in child (button click)
Application	Global	Broadcast across unrelated components
System	Framework level	E.g., init, render, etc.



What are Component Events?

- A component event in Aura is a custom event that bubbles up from a child component to its parent, and optionally to its grandparent if not handled at the immediate level.
- This is useful for notifying higher-level components when something happens in a nested component.
- A component event is fired from an instance of a component. A component event can be handled by the component that fired the event or by a component in the containment hierarchy that receives the event.
- Try to use a component event instead of an application event, if possible. Component events can only be handled by components above them in the containment hierarchy so their usage is more localized to the components that need to know about them.
- An Aura component can be registered to an event using the `<aura:registerEvent>` tag.
- To respond to the event, the `<aura:handler>` tag can be used



Types of Component-Level Events (By Pattern of Use)



Use Case	Event Flow	Event Type
Child -> Parent	Simple Bubbling	Type="component"
Nested Child -> Root Parent	Multi – level bubbling	Type="component"
Reusable child component firing different events	Earth is the only planet of the Solar System that harbors life	Earth is the only planet of the Solar System that harbors life
Dynamic behaviour	Fire events conditionally	Events fired based on attribute/state



Key Characteristics of Component Events

Localized Scope:

Component events are designed for communication within a specific part of the application's component hierarchy.

Containment Hierarchy:

They are primarily used to communicate with parent or child components in the immediate containment relationship.

Registration and Handling:

Component events are registered with a specific name and handled using the `aura:handler` tag, referencing the event and the action to be executed.

`getEvent()` Method:

An instance of a component event is obtained using the `component.getEvent("EventName")`; unlike application events which use `$A.get("e.namespace:EventName")`.

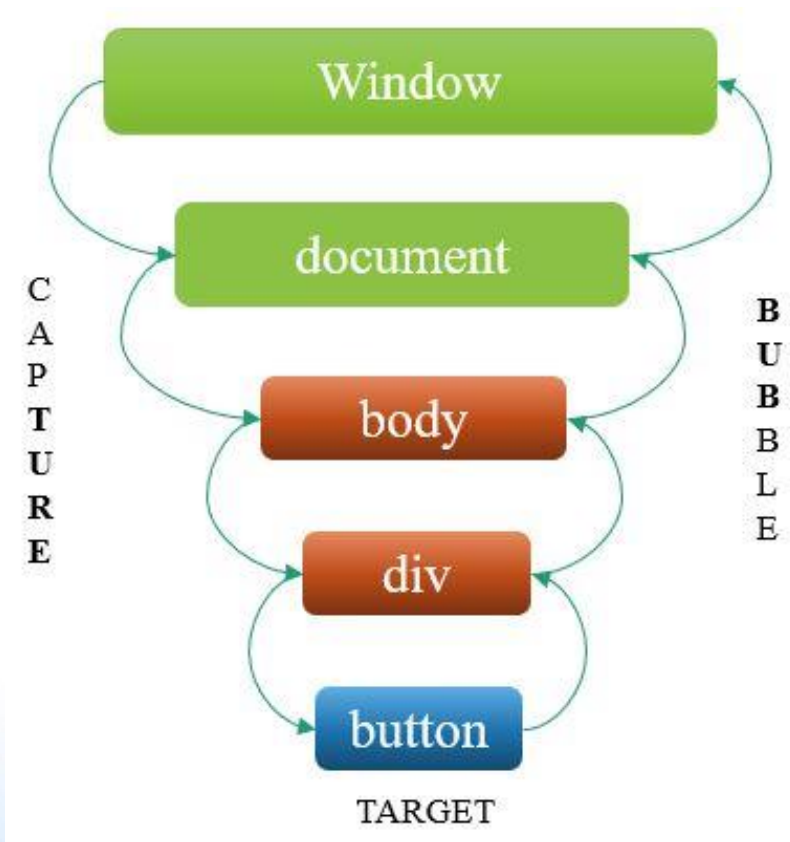
Component Terminology

- **Containment Hierarchy:** refers to a tree of components that has a top-level container as its root component. When an event is triggered, all components within the containment hierarchy are notified
- **Component Event Propagation:** How an event is propagated in the containment hierarchy depends on the phase that is configured in the event handler. There are two types of phases available for component events, namely, Bubble and Capture
 - **Bubble Phase:** The default phase used by a component event is the bubble phase, which behaves in a bottom-up manner. When an event is fired, 1) source component handles the event first. 2) parent component, 3) grandparent, and then each remaining component all the way up until the root component
 - **Capture Phase:** The order of the event propagation in the capture phase behaves in a top-down manner. 1) root component gets to handle the event first, 2) and the propagation traverses down the containment hierarchy until it reaches the source component
- **Stopping Event Propagation:** Event propagation can be terminated any time the stopPropagation method is executed in an event handler



Event Propagation

Event Propagation is a way of communication in your application through concept called bubbling model, , where an event bubbles up through the DOM tree until it's handled or reaches the top.

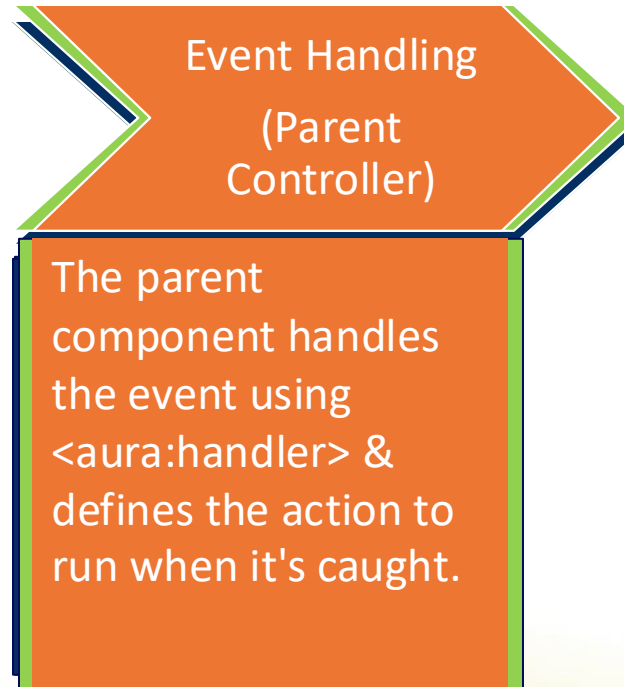


1. Capture Phase: The event is captured from application root to the source component. **[top to bottom]**.

2. Bubble Phase: The event is bubbled up from source component to the application root. **[bottom to top]**

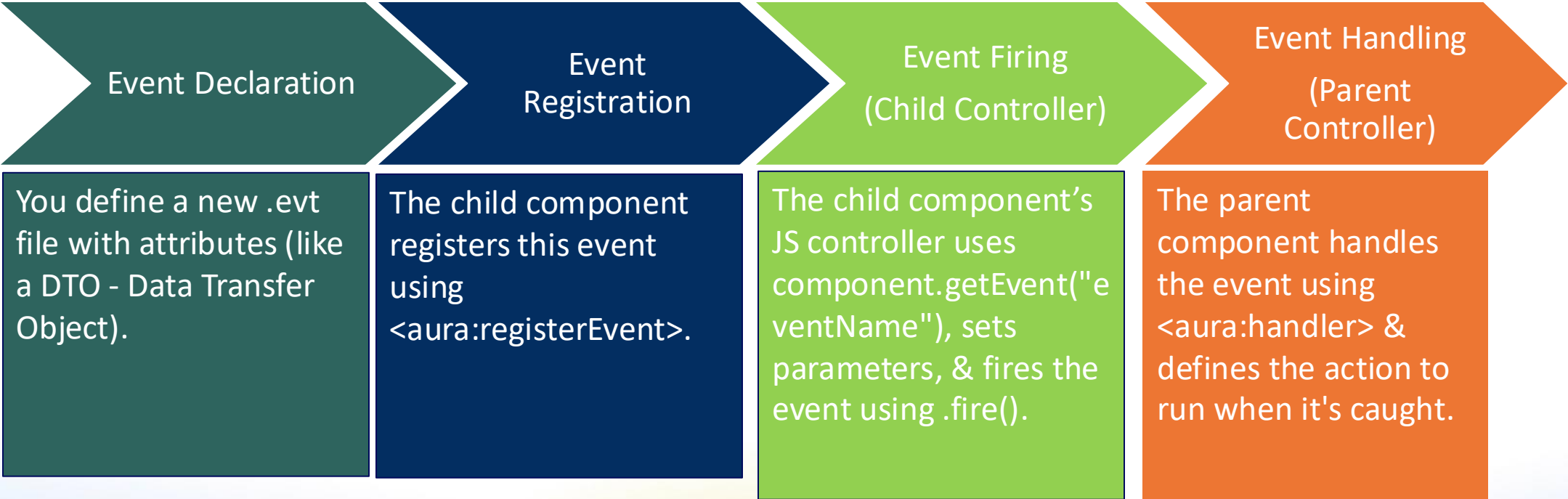
Let's break down the flow

There are 4 major elements for implementing Component level events.



Let's break down the flow

There are 4 major elements for implementing Component level events.



Demo

Component Level Event – Creating
Browser Tiles

salesforce



Use Case



We have a list of students shown in individual tiles. When the user clicks the 'Register' button in a studentTile, the system should notify the parent component (courseRegistration) and update a visible list of registered students. This is a perfect use case for a Component Event.



Component Overview



Trainee Registration

File	Purpose
traineeRegisterEvent.evt	Declares the custom event and its attributes for payload.
traineeTile.cmp	Registers the custom event and then fires the event.
courseRegistration.cmp	Handles the event raised by the traineeTile.



Component vs Application Events





When to use Component Events vs Application Events

Prefer component level events over Application level events

Component Events

Simple communication within a component or its immediate children/parent:

When you need a component to communicate with its immediate parent or children, a component event is usually the best choice.

Localized UI updates:

If you need to update a specific part of the UI based on an action in a child component, component events are appropriate.

Application Events

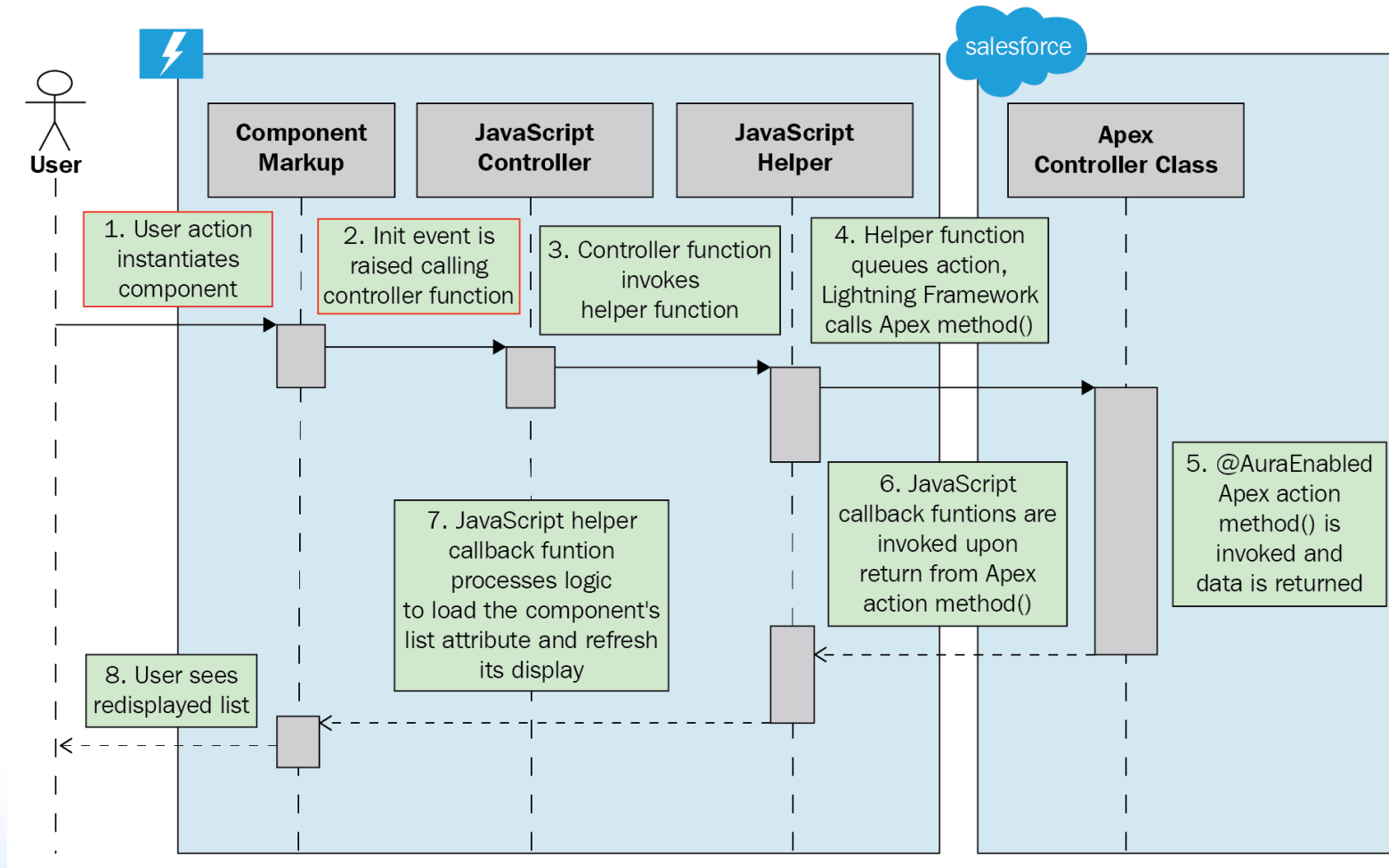
Wider scope:

Application events are for broader communication across the entire application, using a publish-subscribe model.

More complex setup:

They require more setup and are generally used for more global actions like navigation or data updates that need to be reflected throughout the application.

Flow of Execution



thank you

Family Reunion

A stylized illustration of a family reunion signpost in a forest. The signpost is a brown wooden arrow pointing right, mounted on a brown post. It is set against a blue sky with white stars and a green landscape with rolling hills and a line of evergreen trees. The sign itself says "Family Reunion" in a yellow, cursive font.